MATHEMATICS

AGENDA

- What is an Equation?
- Linear equation
- Different types of equations
- 2D **VS** 3D
- Calculs
- Linear Algebra

CI]NSTANT

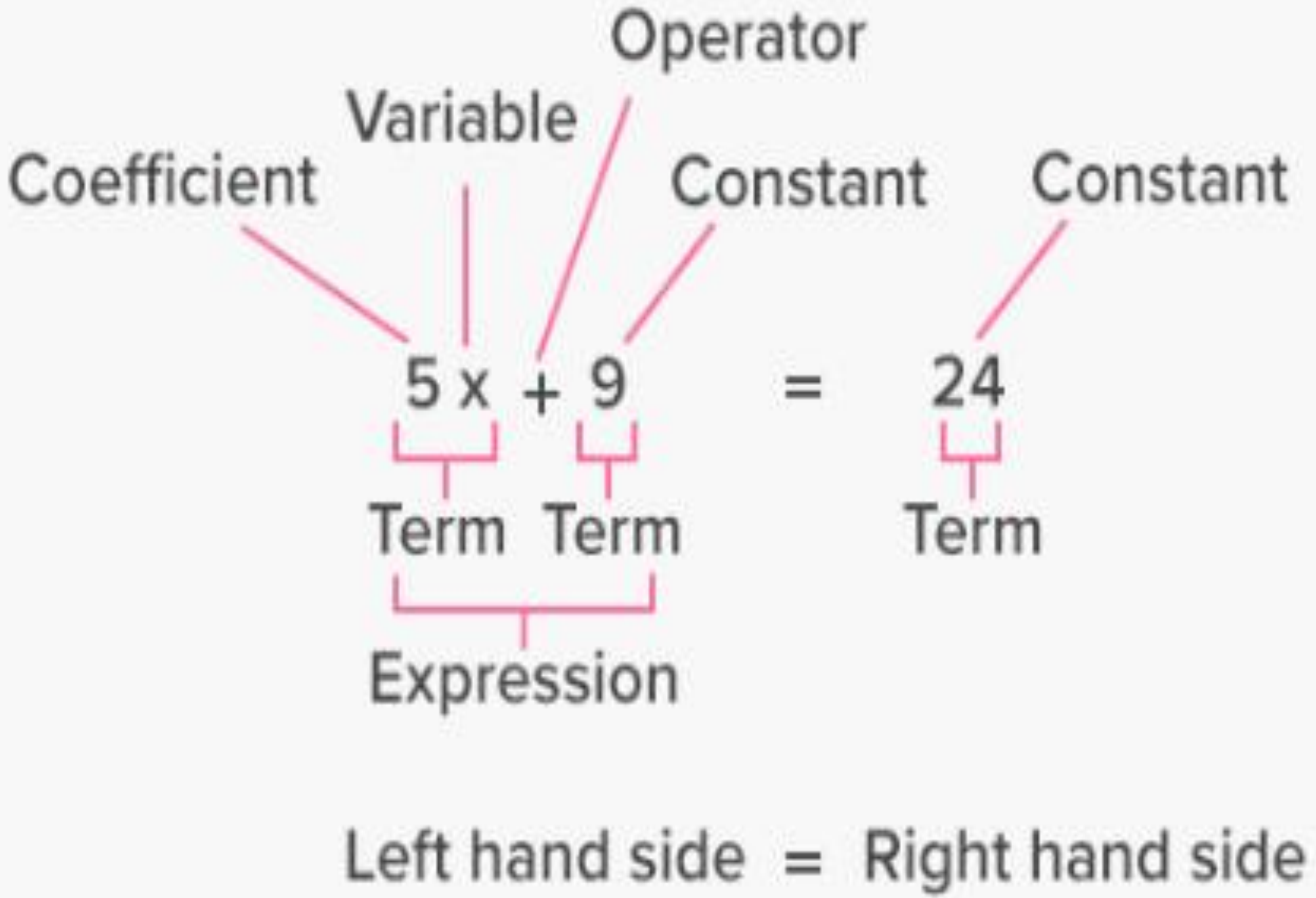
What is an equation?

Equal = the two sides must be equal, so equation generally puts a mathematical structure equal to 0 or equal to Y or some other variable

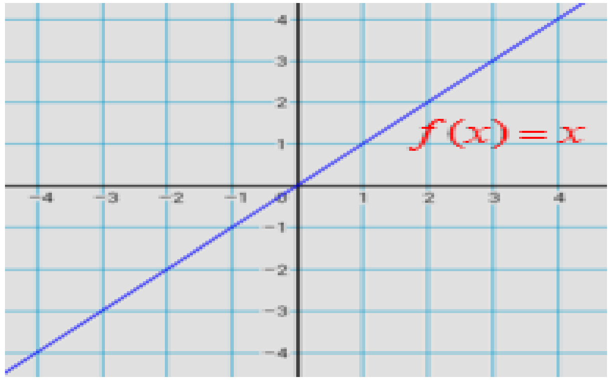
The beginning letters of the alphabet a, b, c, etc. are typically used to denote constants, while the letters x, y, z , are typically used to denote variables. For example, if we write

$$y = ax^2 + bx + c,$$

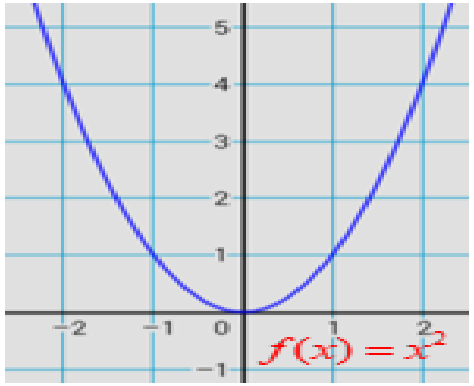
we mean that a, b, c are constants (i.e. fixed numbers), and that x and y are variables.


Operator
Variable
Coefficient
Constant
Constant
5 x + 9 = 24
Term
Term
Term
Expression
Left hand side = Right hand side

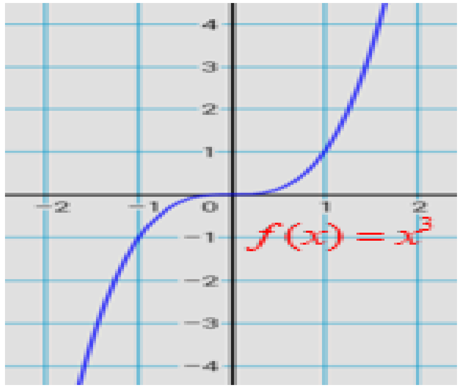
Parent Functions



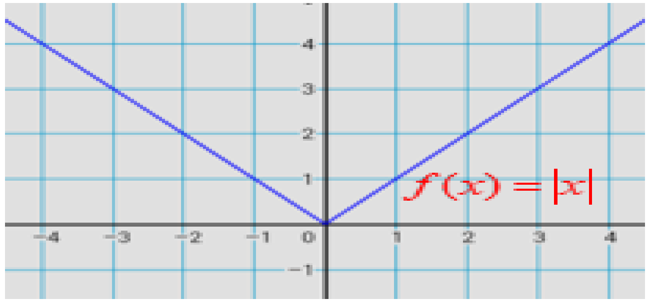
Linear



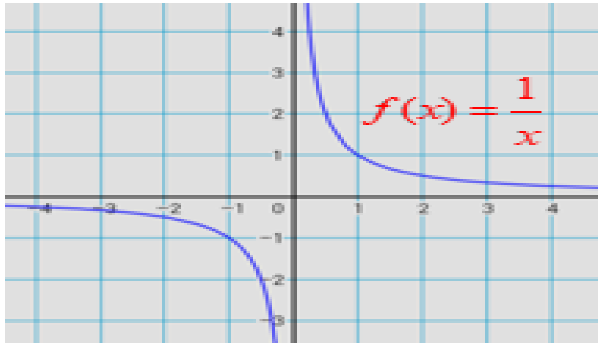
Quadratic



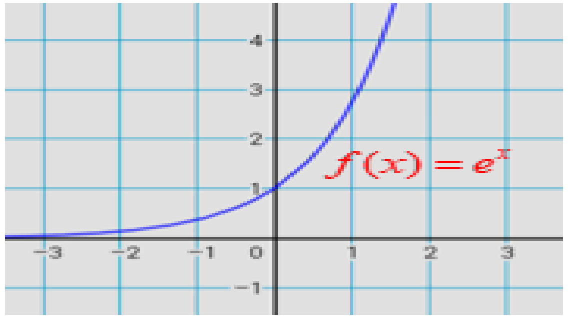
Cubic



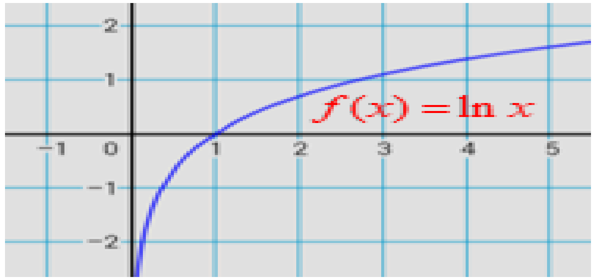
Absolute



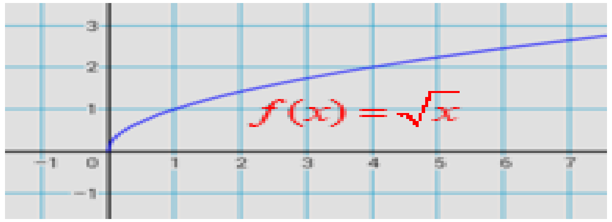
Reciprocal



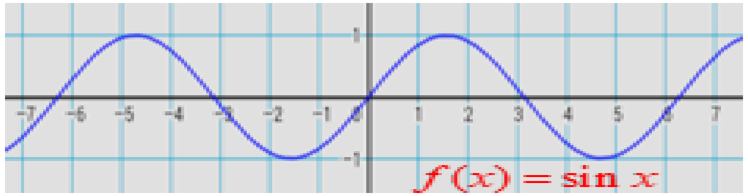
Exponential



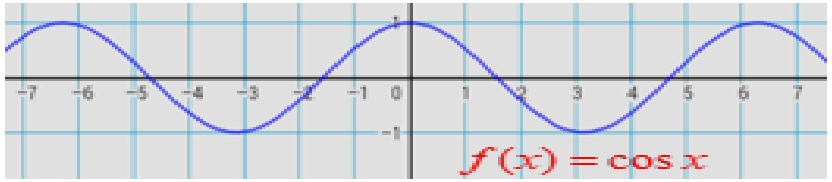
Logarithmic



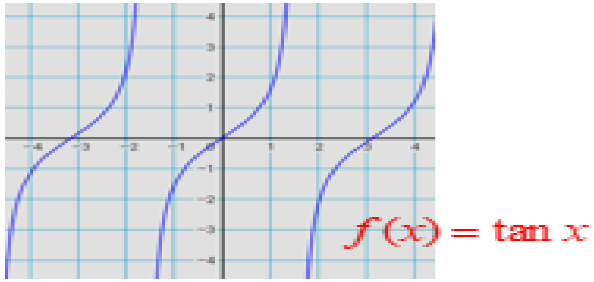
Square Root



Sine



Cosine



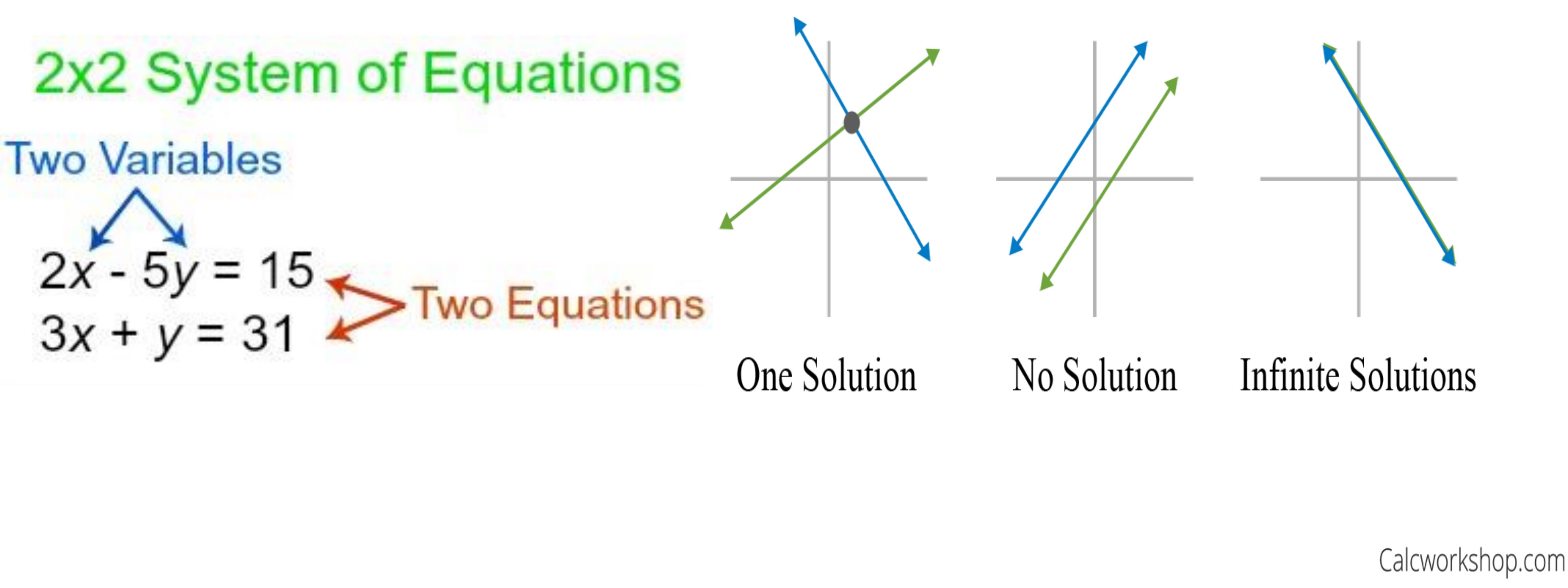
Tangent

Definition

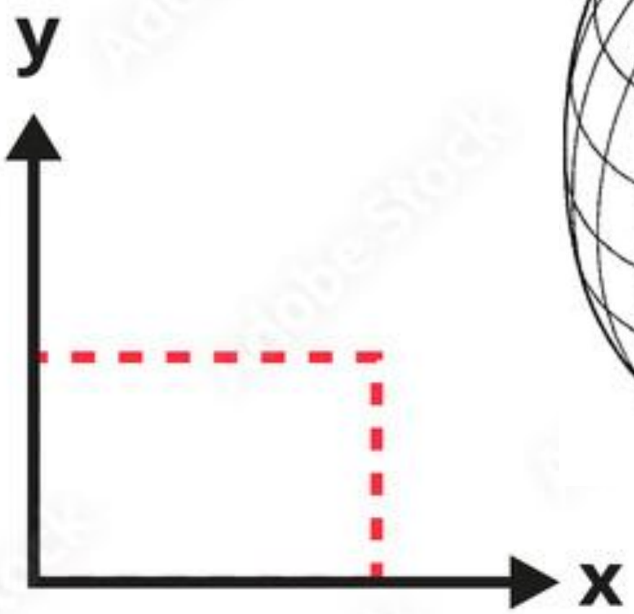
System of Linear Equations — **A collection of linear equations, where the number of equations matches the number of variables, allowing for an algebraic solution.**

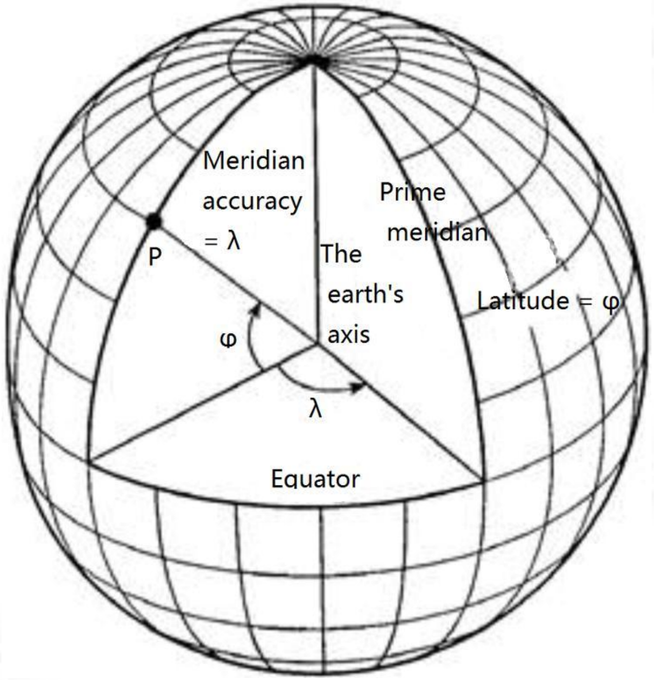
Examples of Linear Systems

2-variable	3-variable
$y = 3x + 1$	$y = 2x + 3z + 10$
$y = -2x + 7$	$y = -4x + 7z - 8$
	$y = -3x + 5z + 12$

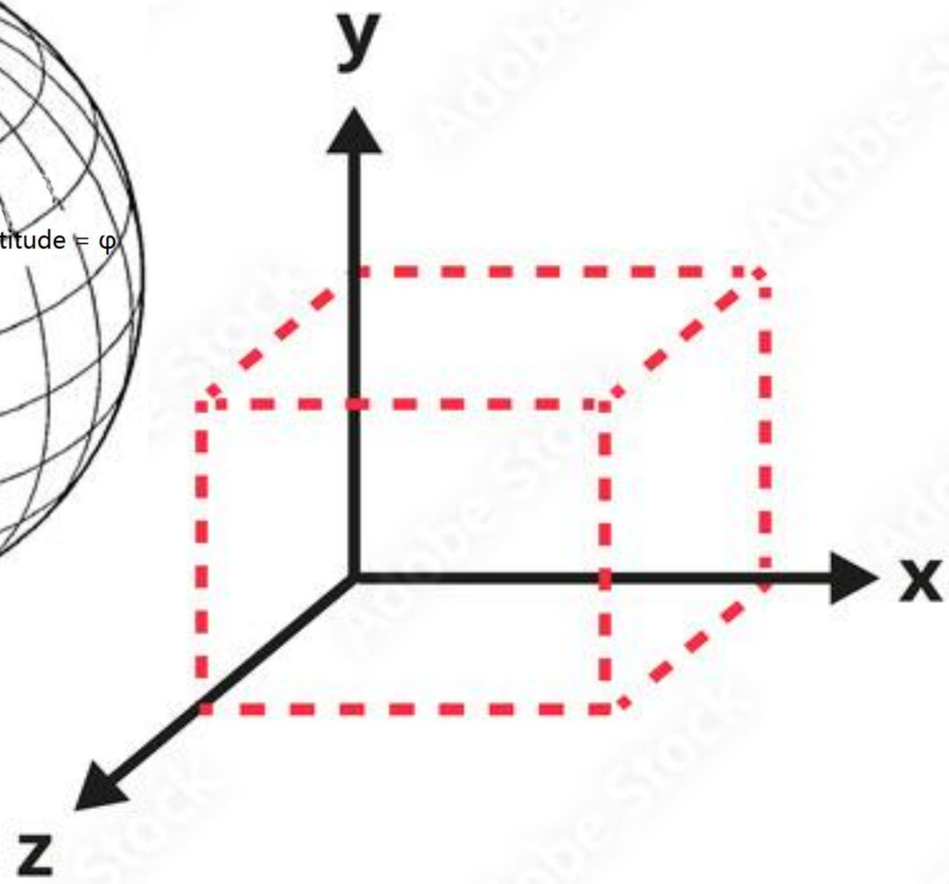

2x2 System of Equations
Two Variables
2x - 5y = 15
3x + y = 31
Two Equations
One Solution
No Solution
Infinite Solutions
Calcworkshop.com

2D Coordinate System



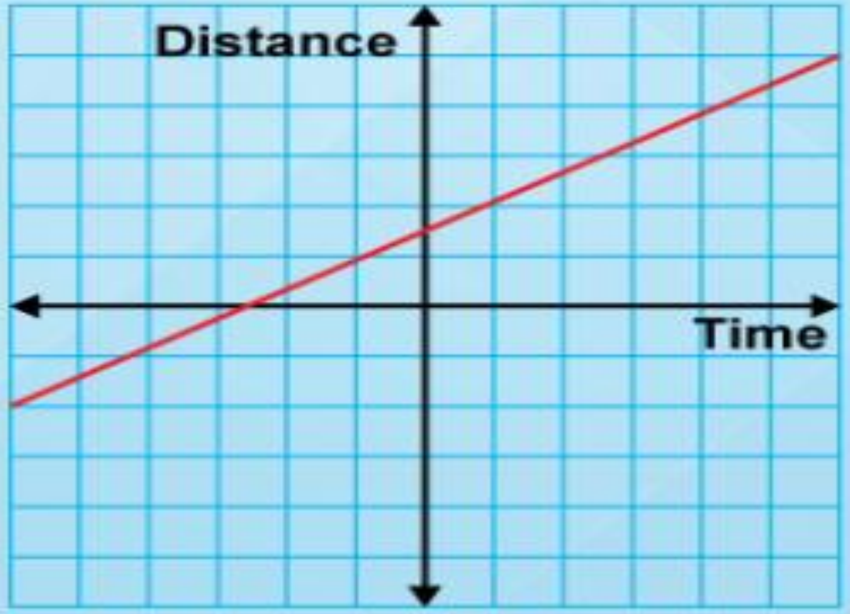


3D Coordinate System



Definition

Rate of Change Slope thought of as the rate at which the dependent variable changes relative to the independent variable.



$$R = \frac{\Delta y}{\Delta x}$$

Speed is the rate of change of distance over time.

WHAT IS LINEAR ALGEBRA?

❑ Definition:

- Linear Algebra is the branch of mathematics focused on vectors, matrices, and linear transformations.

- Forms the backbone of machine learning algorithms and neural networks.

❑ Linear Equation:

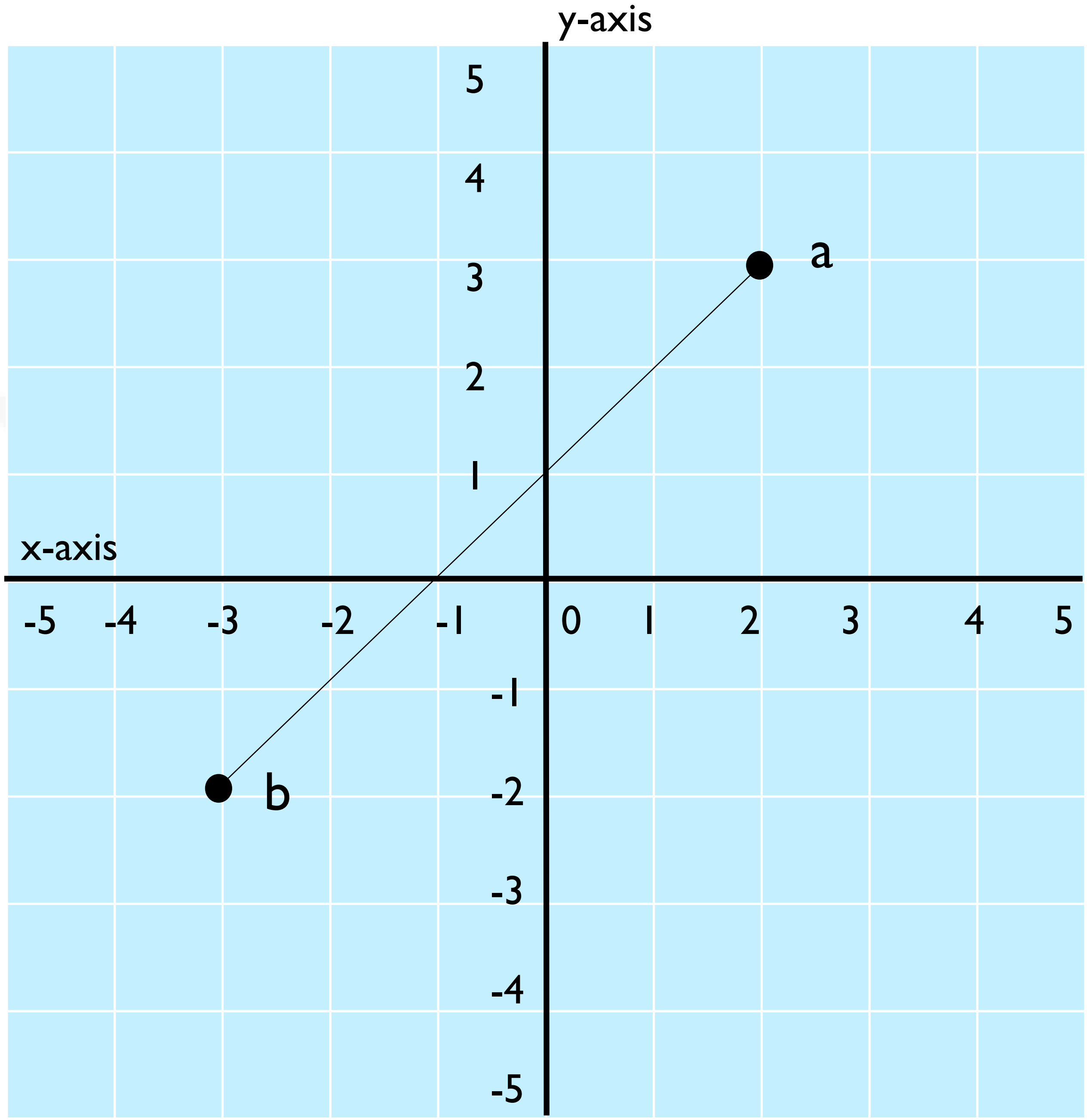
- An equation involving linear expressions of variables.

❑ Linear Equation:

$$a = (2, 3)$$
$$b = (-3, -2)$$

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\frac{y - 3}{x - 2} = \frac{-2 - 3}{-3 - 2}$$



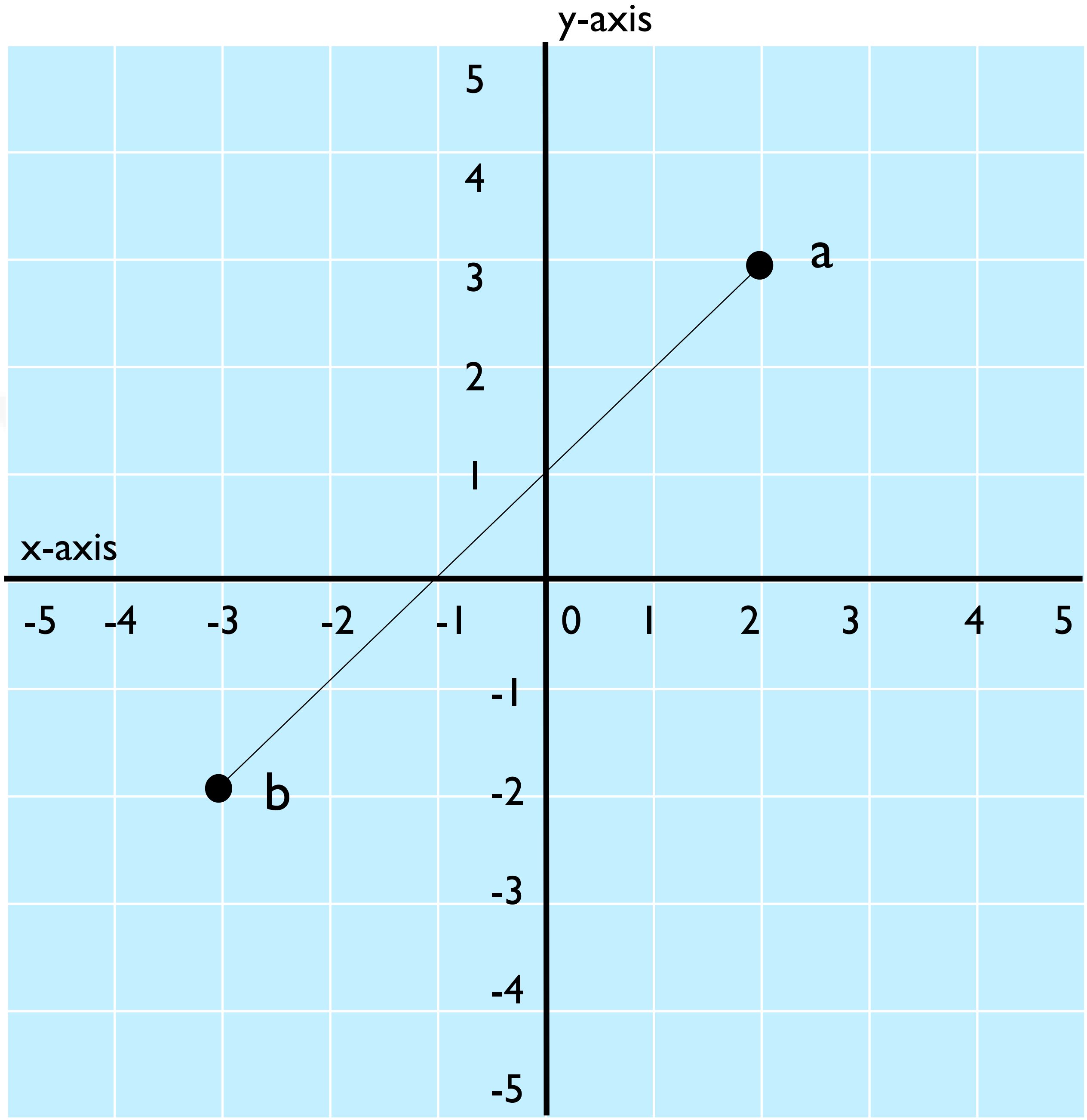
Linear Equation:

$$\frac{y-3}{x-2} = \frac{-5}{-5}$$

$$\frac{y-3}{x-2} = 1$$

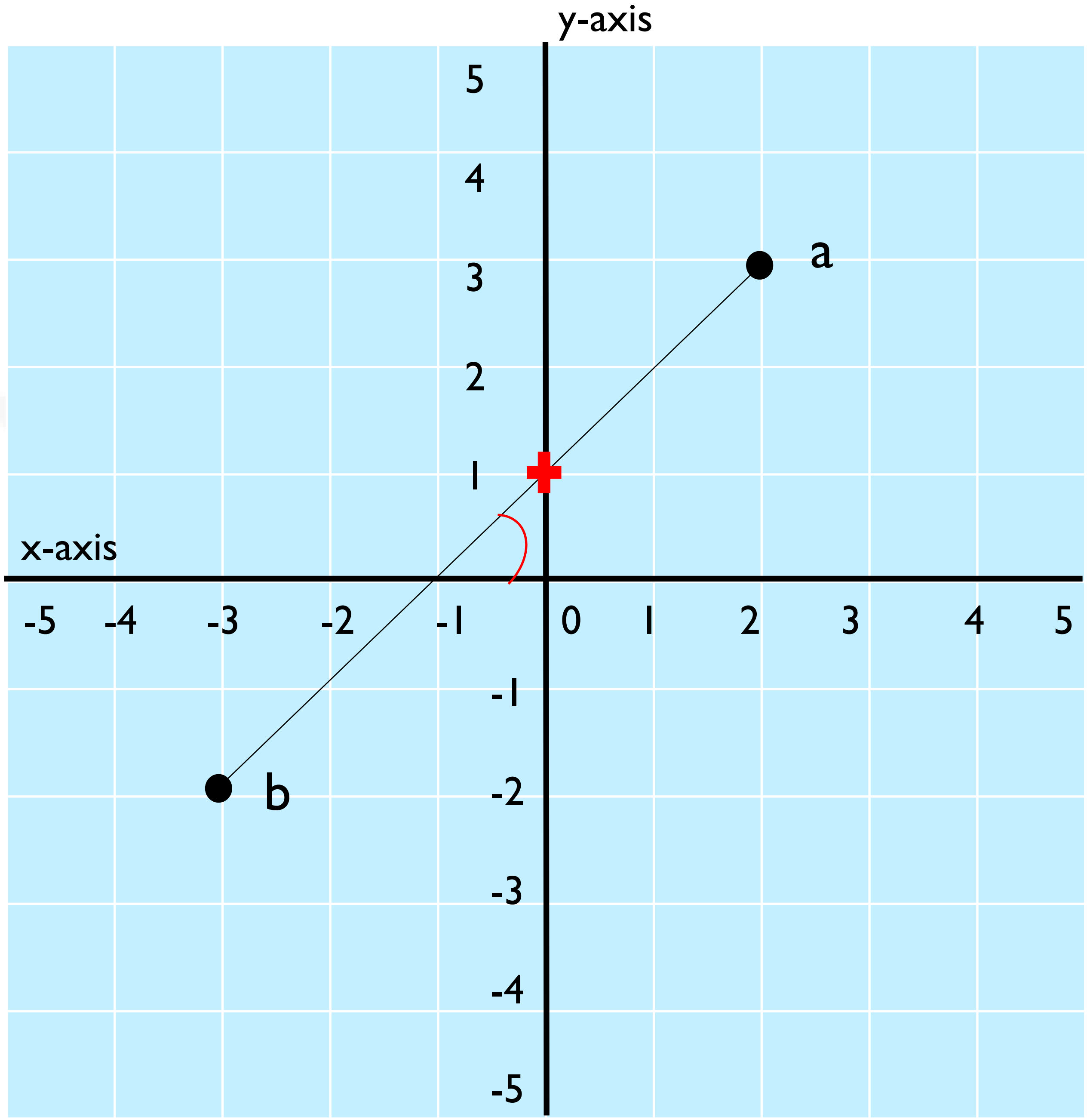
$$y-3 = x-2$$

$$y = x+1$$



- **Linear Equation:**

$$y = x + 1$$

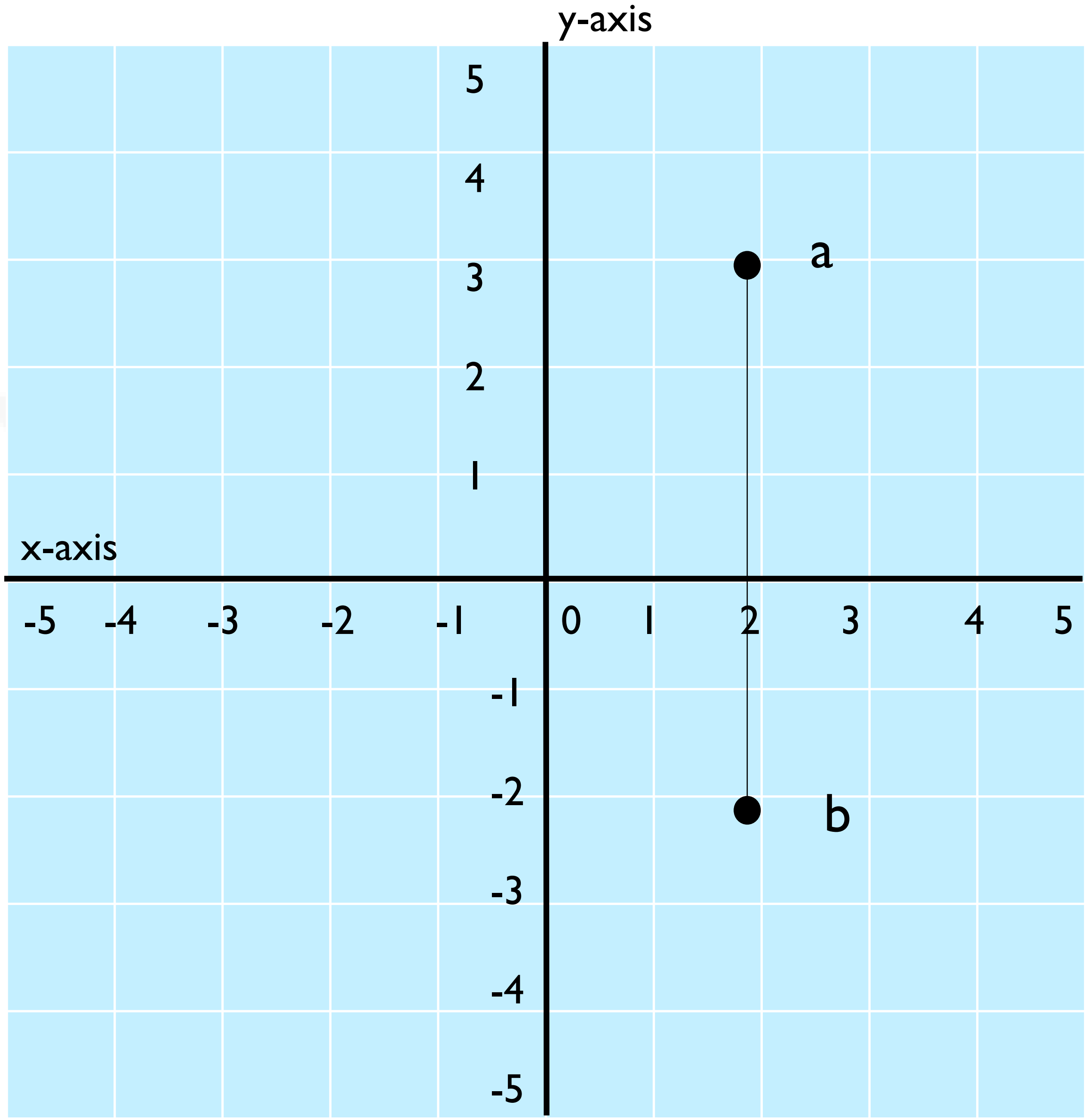


❑ Linear Equation:

$$a = (2, 3)$$
$$b = (2, -2)$$

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\frac{y - 3}{x - 2} = \frac{-2 - 3}{2 - 2}$$



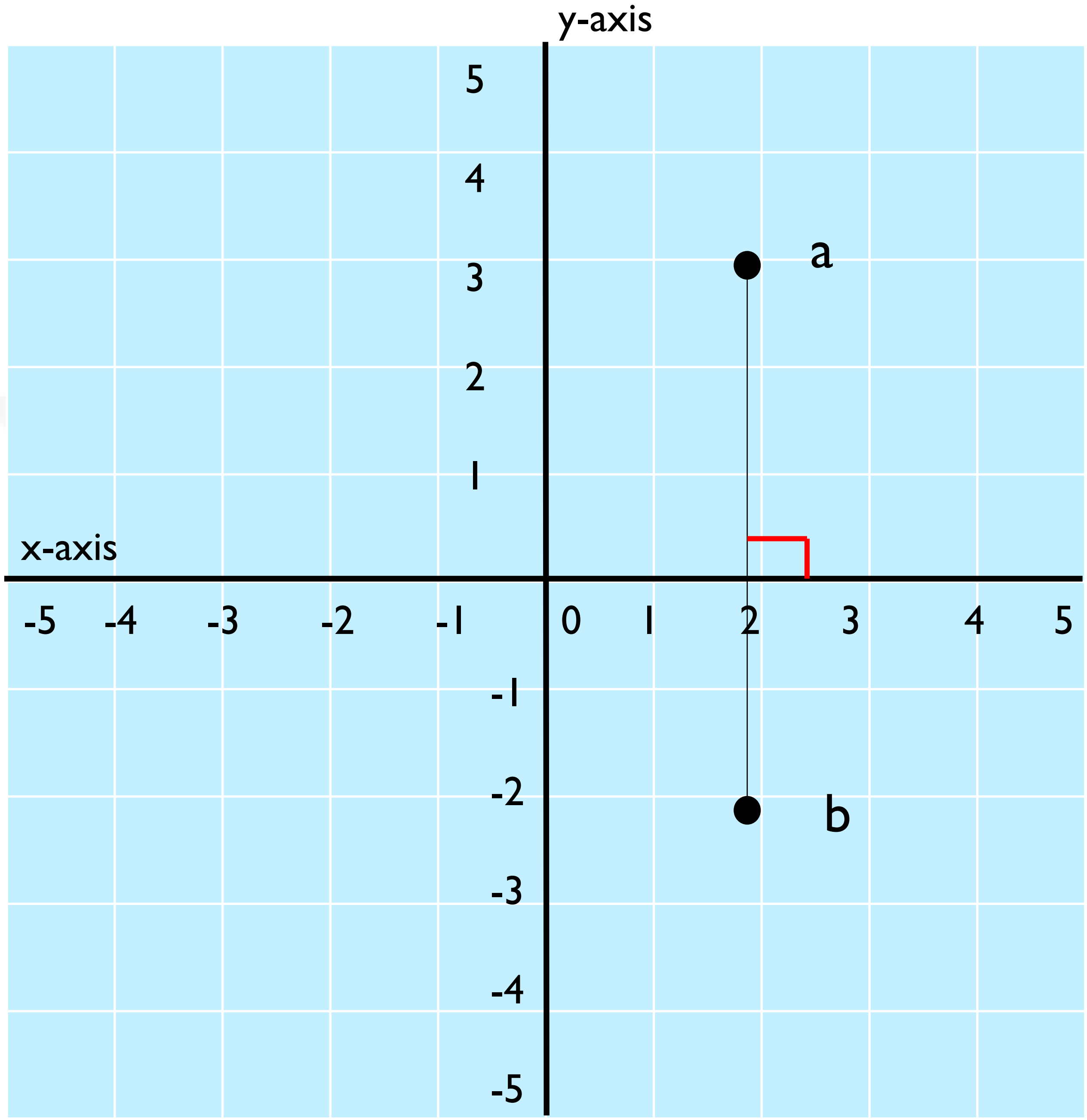
❑ Linear Equation:

$$\frac{y-3}{x-2} = \frac{-5}{0}$$

$$0 = -5x + 10$$

$$5x = 10$$

$$x = 2$$

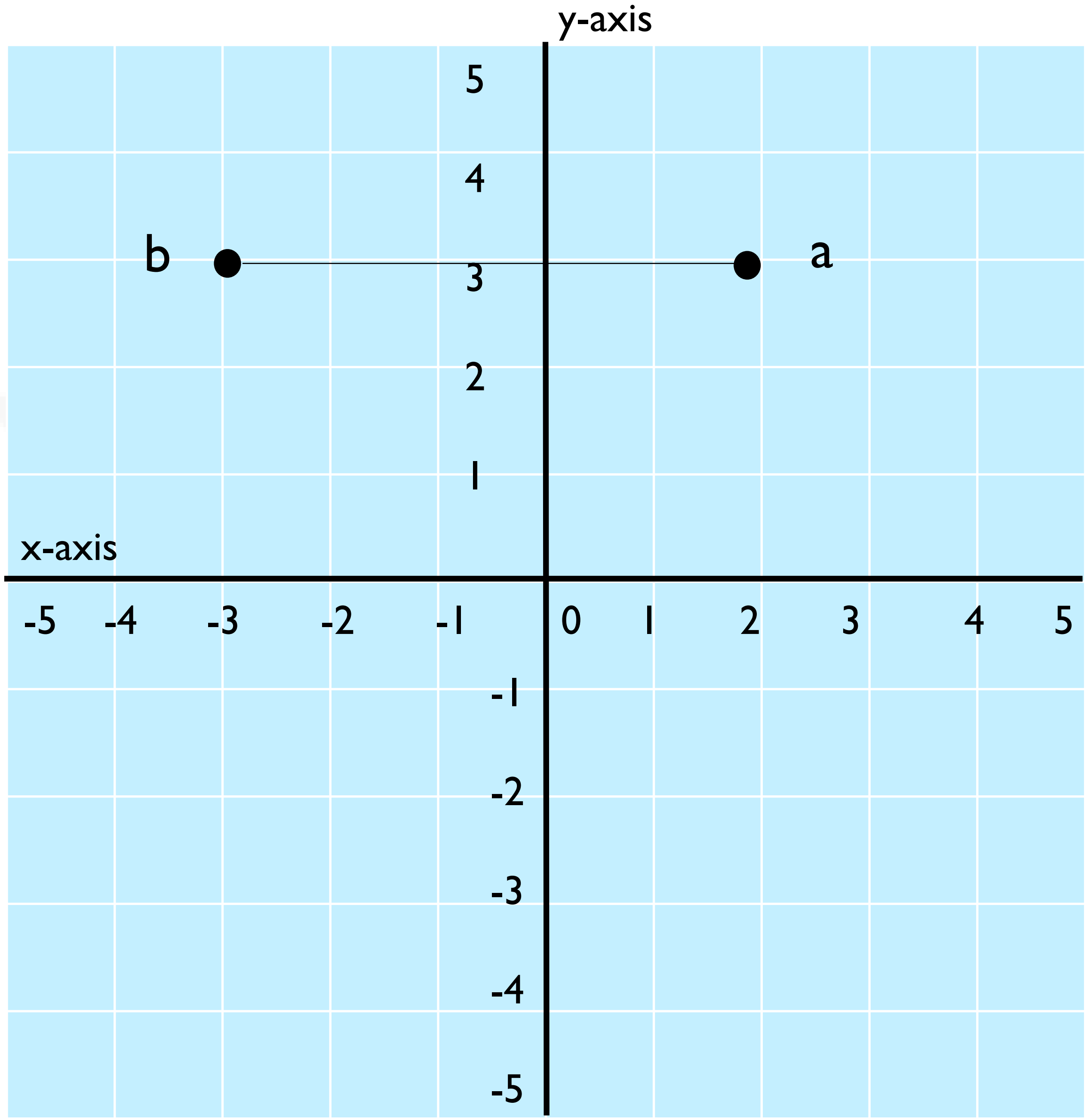


❑ Linear Equation:

$$a = (2, 3)$$
$$b = (-3, 3)$$

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\frac{y - 3}{x - 2} = \frac{3 - 3}{-3 - 2}$$



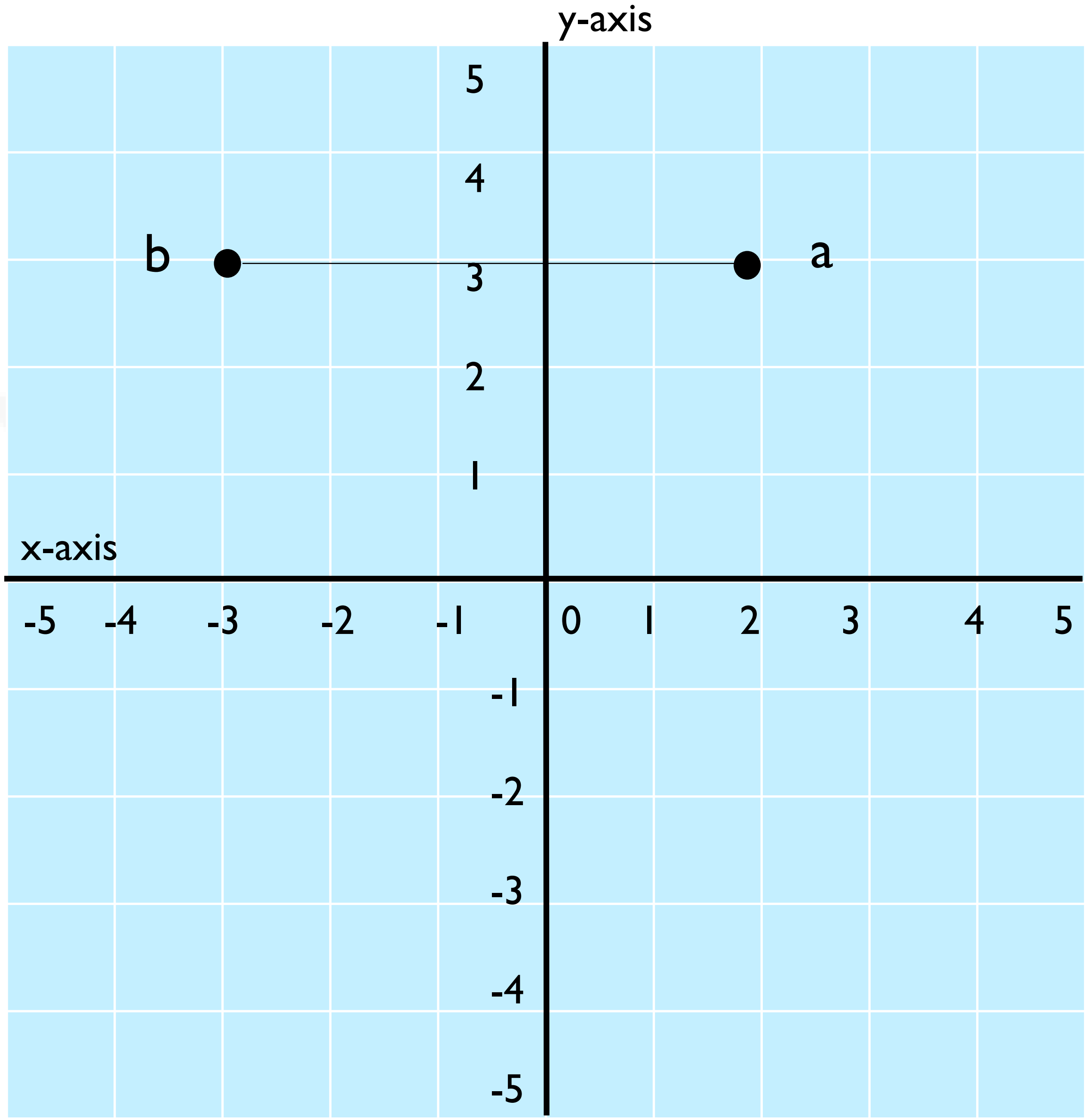
❑ Linear Equation:

$$\frac{y-3}{x-2}=\frac{0}{-5}$$

$$\frac{y-3}{x-2}=0$$

$$y-3=0$$

$$y=3$$

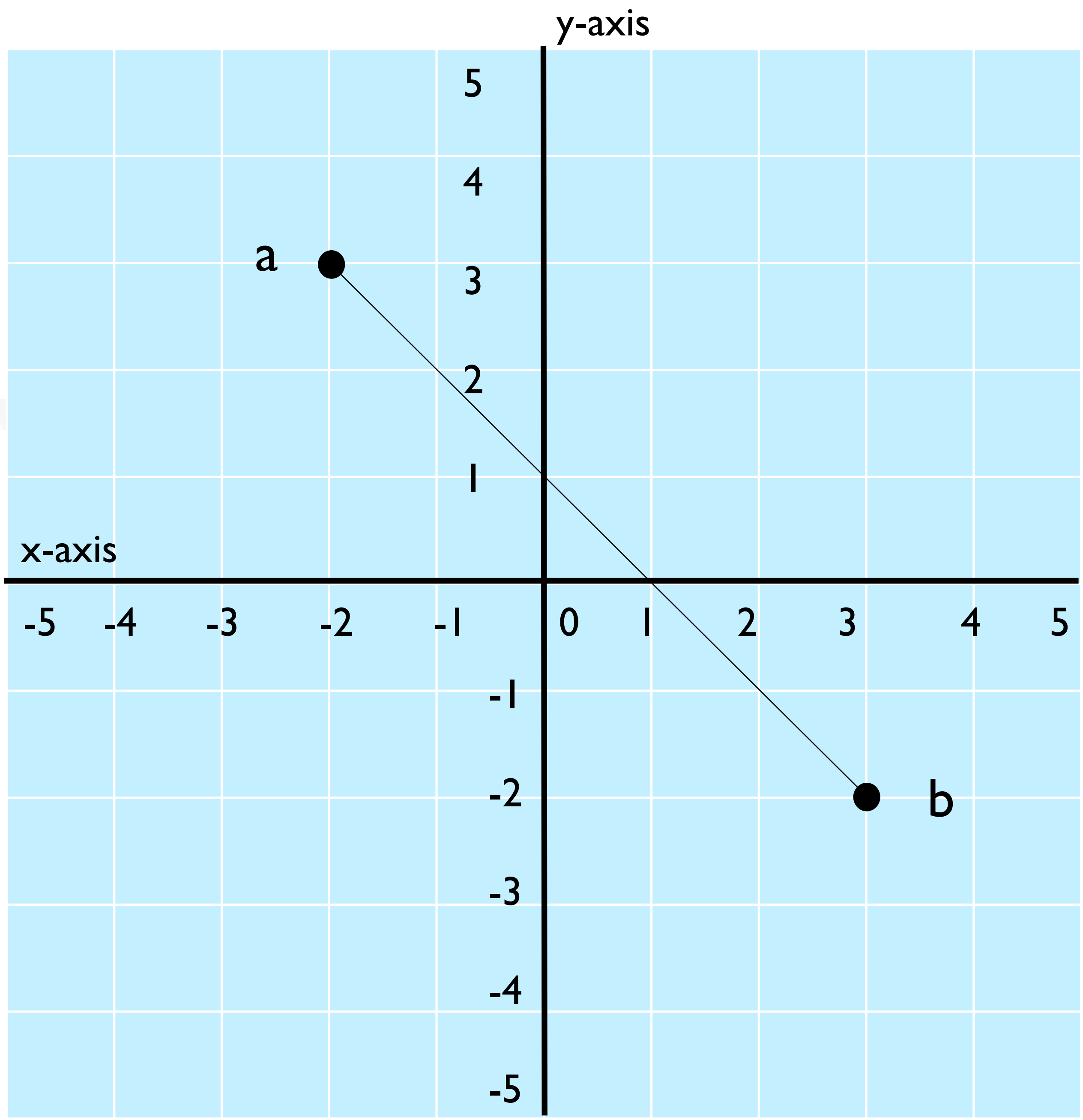


❑ Linear Equation:

$$a = (-2, 3)$$
$$b = (3, -2)$$

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\frac{y - 3}{x + 2} = \frac{-2 - 3}{3 + 2}$$



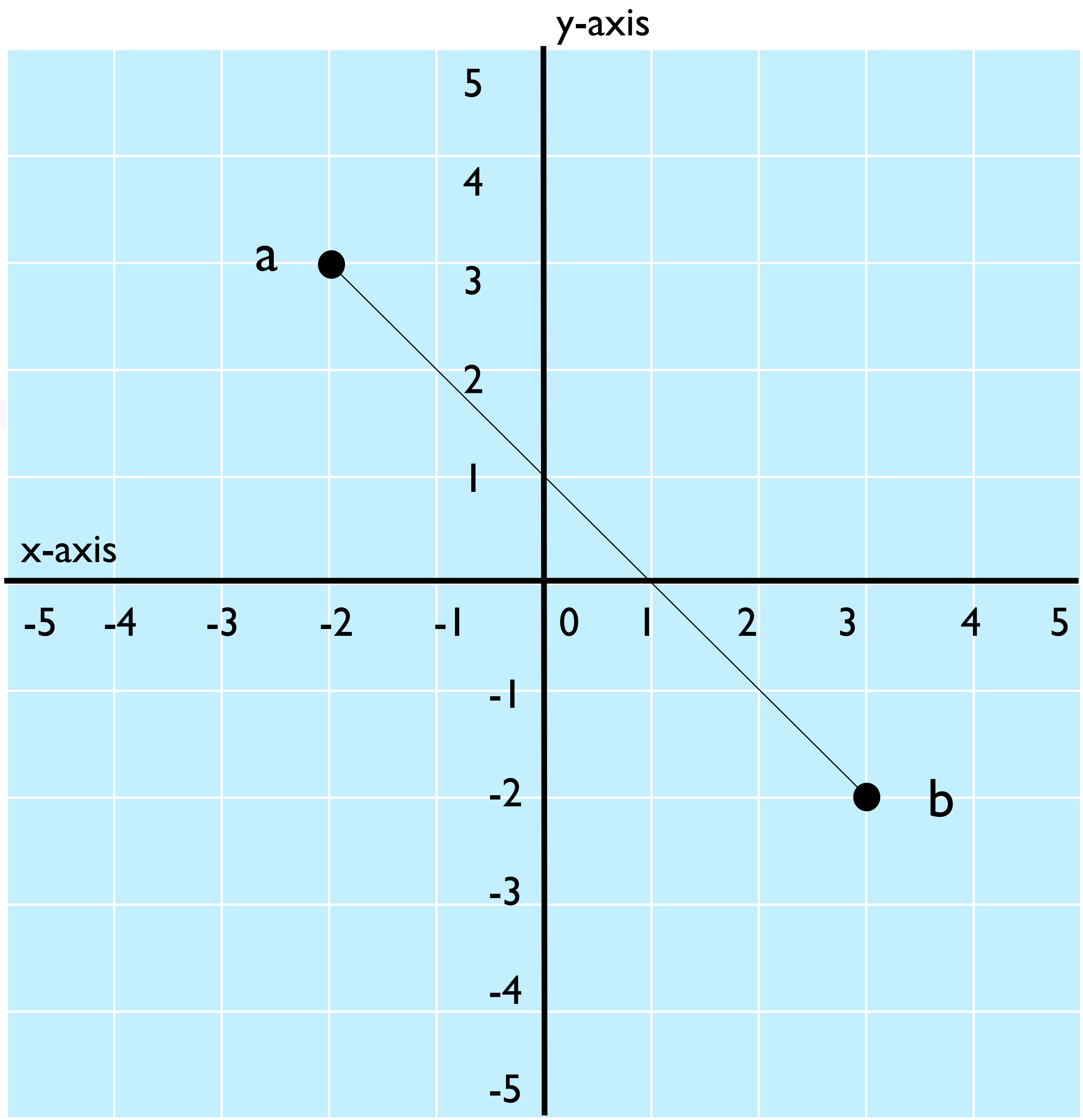
❑ Linear Equation:

$$\frac{y-3}{x+2} = \frac{-5}{5}$$

$$\frac{y-3}{x+2} = -1$$

$$y - 3 = -x - 2$$

$$y = -x + 1$$



VECTORS

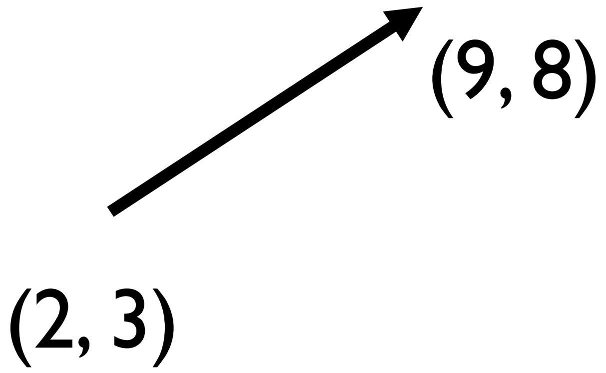
❑ Definition:

- A vector is a quantity with both magnitude and direction, represented as an ordered list of numbers.

$$V = (9,8) - (2,3)$$

$$V = (7,5)$$

$$V = \begin{vmatrix} 7 \\ 5 \end{vmatrix}$$



Operations on Vectors:

- Addition

$$V_1 = (3, 4)$$

$$V_2 = (7, 5)$$

$$V_3 = V_1 + V_2$$
$$V_3 = (10, 9)$$

CIJNSTANT

❑ Operations on Vectors:

- Subtraction

$$V_1 = (3, 4)$$

$$V_2 = (7, 5)$$

$$V_3 = V_2 - V_1$$
$$V_3 = (4, 1)$$

❑ Operations on Vectors:

- Multiplication

$$V_1 = (3, 4)$$

$$V_2 = (7, 5)$$

$$V_3 = V_2 * V_1$$
$$V_3 = (21, 20)$$

CIJNSTANT

❑ Operations on Vectors:

- Scaler Multiplication

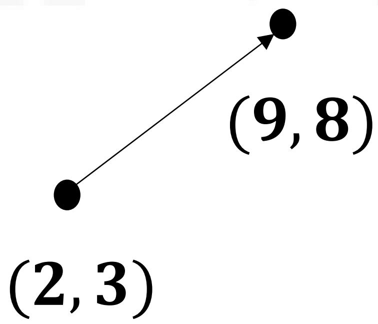
$$V = (3, 4)$$

$$3V = 3 * V_1$$
$$3V = (9, 12)$$

❑ Operations on Vectors:

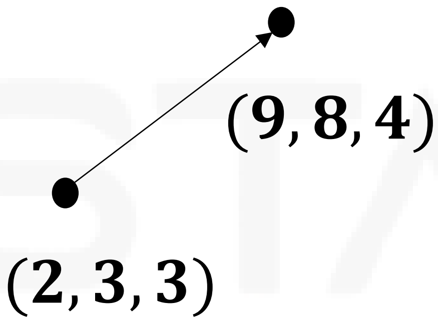
$$V = \begin{bmatrix} 9 \\ 8 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$V = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$$



❑ Operations on Vectors:

$$V = \begin{bmatrix} 9 \\ 8 \\ 4 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}$$



$$V = \begin{bmatrix} 7 \\ 5 \\ 1 \end{bmatrix}$$